

1 - (4) Quantitative sales forecasting (Four Quarter Moving Average & Extrapolation)



Learning Outcome

To be able to apply and evaluate quantitative techniques for sales forecasting, including four-period moving averages.

To Know



- Calculate moving averages, including four-period and four-quarter moving averages, to identify trends.
- Interpret scatter graphs and draw a line of best fit.
- Discuss the advantages and limitations of quantitative sales forecasting techniques.
- Discuss when quantitative methods may be unreliable due to external factors or changing business conditions.

Why use Four Quarter Moving Averages?

Many businesses will wish to **smooth data over the four quarters of the year.**

This is because there could be significant fluctuations due to the **seasonal nature of the business.**

In this case they will need to use **four quarter moving averages.**

Important: Centring the Average

If they do this unlike the 3 period moving average there is **NO MIDDLE POINT** to plot the moving average figure (TREND)

Therefore a process of **CENTRING** is used as shown below.

Date	Sales	4 point total	8 point total	4 point centred average	Variation (Actual - Trend)
2018: Q1	400				
2018: Q2	500				
		2570			
2018: Q3	770		5340	667.5	102.5
		2770			
2018: Q4	900		5740	717.5	182.5
		2970			
2019: Q1	600		6270	783.75	-183.75
		3300			
2019: Q2	700		7200	900	-200
		3900			
2019: Q3	1,100		7800	975	125
		3900			
2019: Q4	1,500		7900	987.5	512.5
		4000			
2020: Q1	600				
2020: Q2	800				
2020: Q3	1200				
2020: Q4	1400				

Calculate the following:

Average Variance from the Trend =

Cyclical variation - only takes into account variations for each cycle point (in this case - Q1, Q2, Q3 and Q4)

E.g. Q1 - add together the Q1 variations only and divide by 2

Calculate the cyclical variation for Q1 =

Q2 =

Q3 =

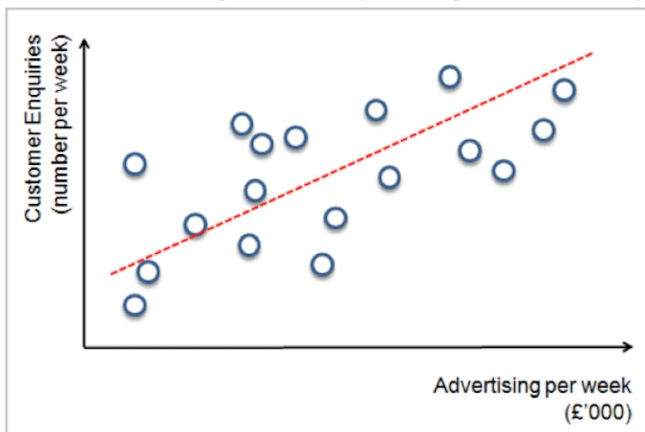
Q4 =

Correlation

Correlation is another method of sales forecasting. Correlation looks at the strength of a relationship between two variables.

For marketing, it might be useful to know that there is a predictable relationship between sales and factors such as advertising, weather, consumer income etc.

Correlation is usually measured by using a scatter diagram, on which data points are plotted. For example, a data point might measure the number of customer enquiries that are generated per week (x-axis) against the amount spent on advertising (y-axis). This is illustrated below:



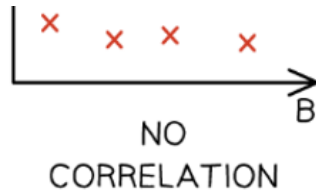
The line of best fit indicates the **strength of the correlation**.

Strong correlation means that there is little room between the data points and the line.

Weak correlation means that the data points are spread quite wide and far away from the line of best fit.

If the data suggests strong correlation, then the relationship might be used to make marketing predictions.





- **No correlation** means there is **no connection** between the two variables
 - It is not possible to identify **a line of best fit**

What are benefits and limitations of Quantitative Sales Forecasting?

Practice question

Extract D

Live the Adventure Ltd

Live the Adventure Ltd is a family business founded in 1984 which operates a leading UK activity centre in Shropshire. It offers an extensive range of activities, including archery, caving, climbing, horse riding and paintballing. The centre has a well-qualified permanent team of staff, supported by a number of seasonal employees.

Since 2010 Live the Adventure Ltd has provided expedition holidays at its other centre in Nepal, which specialises in kayaking, rafting and trekking holidays. Live the Adventure Ltd is considering offering winter activity holidays in Europe, such as skiing and snowboarding, to widen its customer base.

The quarterly sales for Live the Adventure Ltd, 2015-2017 are on the next page.

Quarterly sales for Live the Adventure Ltd, 2015-2017

Year and *Quarter	Sales in £'000s	4 Quarter Total	8 Period Total	4 Quarter Moving Average (Trend)	Variation
2015 Q1	40				
2015 Q2	70				
		240			
2015 Q3	80		490	61.25	18.75
		250			
2015 Q4	50		510	63.75	-13.75
		260			
2016 Q1	50		530	66.25	-16.25
		270			
2016 Q2	80		550	68.75	11.25

		280			
2016 Q3	90		570	71.25	18.75
		290			
2016 Q4	60		590	A	B
		300			
2017 Q1	60		610	C	D
		310			
2017 Q2	90		630	78.75	11.25
		320			
2017 Q3	100				
2017 Q4	70				

Live the Adventure Ltd is deciding whether to expand its existing operations or to offer winter activity holidays in Europe in order to grow.

(d) Using the data in Extract D, calculate the values of A, B, C and D and, using other non-financial information, evaluate these two options.

Recommend which **one** of these two options Live the Adventure Ltd should choose in order to grow.

(20)